Matching Compustat Data to the Longitudinal Business Database, 1976-2020

by

Cristina Tello-Trillo
U.S. Census Bureau

Lawrence Schmidt
MIT

Sean Streiff
Econometrica

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Abstract

This paper details the methodology for creating an updated Compustat-Longitudinal Business Database (LBD) bridge, facilitating linkage between company identifiers in Compustat and firm identifiers in the LBD. In addition to data from Compustat, we incorporate historical data on public companies from various public and private sources, including information on executive names. Our methodology involves a series of stages using fuzzy name and address matching, including EIN, telephone number, and industry code matching. Qualified researchers with approved proposals can access this bridge though the Federal Statistical Research Data Centers. The Compustat-SSL bridge serves as a crucial resource for longitudinal studies on U.S. businesses, corporate governance, and executive compensation.

* Tello-Trillo, Cristina, U.S. Census Bureau; Lawrence Schmidt, MIT; Sean Streiff, Econometrica, sstreiff@econometricainc.com. Any opinions and conclusions expressed herein are those of the authors and do not represent the views of the U.S. Census Bureau. The Census Bureau has ensured appropriate access and use of confidential data and has reviewed these results for disclosure avoidance protection (Project 7514232: CBDRB-FY24-CES022-003).

1 Introduction

The integration of comprehensive financial and business datasets is crucial for understanding corporate behavior and its broader economic implications. This paper presents a refined methodology for creating an updated bridge between Compustat data and the Longitudinal Business Database (LBD), covering the period from 1976 to 2020. By linking company identifiers from Compustat with firm identifiers in the LBD, this bridge facilitates enhanced analyses of public companies' financial and operational dynamics over an extensive historical period.

Our approach incorporates a multifaceted data collection process, leveraging public and private sources to compile a rich dataset of public firms. The primary data source, Compustat Fundamentals, provides detailed financial and employment information, which is supplemented with historical company data from Compustat Historical Header Data, Center for Research in Security Prices (CRSP) and SEC filings. These combinations of datasets allow us to track companies consistently over time using unique identifiers such as GVKEY.

Our ultimate goal is to link GVKEY identifier from Compustat to a FIRMID identifier from LBD for every year, however we should note that this link is not always one-to-one. Even though they both identify companies, they serve different purposes based on their data focus. A GVKEY identifies a company (issue, currency, index) in the Capital IQ Compustat database, which focuses on financial data and security issuances. A FIRMID identifies an economic unit as defined by the US Census Bureau: a company is an economic unit comprising one or more establishments under common ownership or control.[1] The latter definition focuses on the operational structure of a business while the former focuses more on the financial aspect of the business.

The relationship between GVKEY and FIRMID can be complex. Although we tried to do one-to-one GVKEY-FIRMID match, this is not always possible. First, we find that is usually the case that a GVKEY corresponds to one of more FIRMID and not the other way around (one FIRMID to more than one GVKEY) as shown in Figure 1.[2] Second, our disambiguation process only allows a FIRMID to belong to one and only one GVKEY each year, this means the scenario with an 'X' in Figure 1 isn't possible.

To achieve accurate matches between Compustat and LBD data, we employed a series of sophisticated fuzzy matching techniques. These techniques involved matching on variables like company name, address, Employer Identification Number (EIN), and industry codes, ensuring a high degree of accuracy in linking records across datasets. The resulting Compustat-

[1] Extracted from the Company Organization Survey Instructions. See Appendix B.

[2] In nearly all cases we examined, a single GVKEY corresponds to one or more FIRMIDs. However, we rarely encountered situations where a single FIRMID linked to multiple GVKEYs.

LBD bridge, accessible within the Integrated Research Environment (IRE) data warehouse, represents a significant resource for researchers conducting longitudinal studies on U.S. businesses.[3]

Our methodology also addresses critical challenges such as data standardization, de-duplication, and refinement of matching criteria. By systematically cleaning and organizing the data, we ensure the reliability of our matches, focusing on financially active companies within the United States. Furthermore, the integration of executive information data through probabilistic name-matching techniques enhances the scope and depth of our analysis, providing future valuable insights into corporate governance and executive compensation.

This paper details each stage of our matching process, from initial data preparation to the final de-duplication of matches. We present comprehensive statistics on matching success rates, illustrating the robustness of our approach. Ultimately, the Compustat-LBD bridge offers a powerful tool for economic researchers, enabling more precise and comprehensive analyses of corporate behavior over the past four decades.

2 Data

2.1 Public-Firm Data

Our primary source for company identifiers and financial information was Compustat Fundamentals data.[4] This dataset, spanning the period 1976-2020, provided details on individual companies including firm name, address, Employer Identification Number (EIN), and various financial metrics. We used the unique identifier GVKEY to track companies throughout the analysis.

Compustat Fundamentals served three key purposes in our study. First, it offered the universe of company-year pairs (GVKEY-year) needed for matching companies across databases. Second, it provided employment and financial data to define "active" companies in Compustat. Companies were considered active if they had a positive value in employment, revenue, price, or assets for a given year. Finally, Compustat Fundamentals offered four quantitative variables (employment, revenue, assets, market value) that were crucial for evaluating the effectiveness of the matching process. One key metric we used was the proportion of the total value for each variable within Compustat that successfully matched to the Linked Business Database (LBD).

[3]While the bridge is available in the data warehouse, researchers must obtain access to the Compustat data separately.

[4]Downloaded from WRDS on June 1, 2021.

In addition to Compustat Fundamentals, which provided the most recent data on company information, we gathered historical data from additional sources: the Legacy CRSP/Compustat Historical Header Data provided historical details on company identifiers, names, locations, and security information for each company-year combination. We complement this information with additional historic company names and address from SEC filings, 10-Ks, where we used Compact Disclosure (Compact D/SEC) from 1992-2006 and Edgar Data from 2009 onwards. Finally, CRSP Data provided access to additional historical company names. This multifaceted approach ensured a thorough understanding of the public firms included in the study.

From these datasets, we extract six "matching variables" to link Public-Firm data to Business Register (BR) and LBD data:

1. Employer Identification Number (EIN). Obtained from Fundamentals, historical headers, and Edgar,
2. Company Name (obtained from all 5 sources)
3. Street Address (obtained from Fundamentals, historical headers, Compact Disclosure, and Edgar – all sources except CRSP),
4. City, State, and ZIP Code (combined, obtained from all sources except CRSP),
5. Telephone Number (obtained from all sources except CRSP),
6. NAICS, at the 4-digit level[5] (obtained from Fundamentals only).

2.1.1 Public-Firm Data Cleaning and Refinements

To prepare the data for matching, we undertook initial cleaning procedures that preserved the overall number of observations and unique companies. Firstly, we established the observation year using "fiscal year" data, or "data year" if fiscal year was absent. Company names, addresses, and zip codes underwent cleaning and standardization to enhance matching accuracy, involving tasks such as removing hyphens from EINs and refining zip codes. Text substitution algorithms and SAS DQ Match code at sensitivity 95 were then applied to standardize variables like company names, addresses, and cities for matching preparation.

Subsequently, we streamlined the Compustat dataset to focus on our targeted active company cohort. This involved de-duplicating data based on GVKEY and year, prioritizing non-financial service records. For further refinement, we excluded observations with business

[5]The Fundamentals file has NAICS data at the 6-digit level for most GVKEYs, but we used 4 digits for matching to the BR.

descriptions containing keywords like "merger" or "acquired by" if they duplicated based on EIN and year. Additionally, observations were removed if their CUSIP indicated subsidiary status and duplicated based on EIN and year. Among remaining duplicates based on EIN and year, only the instance with the highest annual stock price was retained.

Records were retained exclusively for companies located within the United States and excluded entities classified with a 3-digit NAICS code of 525 (indicating involvement in financial instruments). We included records for companies with stko values of 0 or 3, representing publicly traded companies on major exchanges and other platforms respectively, while excluding subsidiaries (stko=1,2) and companies involved in leveraged buyouts (stko=4).

Finally, we defined "active" companies as those demonstrating positive values for at least one of the following: employment, revenue, stock price, or total assets in a given year. Only GVKEY-year pairs meeting these criteria were retained, ensuring our analysis focuses on relevant, financially active companies operating within the United States.

2.2 Business Register Data Preparation

This section describes the approach taken to prepare Business Register (BR) data for matching with Compustat information.

Previous matching algorithms created a single, consolidated BR file for each year (1976-2020), encompassing all relevant BR variables (totaling 45 files, one file per year). In this project, we opted for a more efficient approach, generating only six distinct BR files. Each file focuses on a specific BR variable crucial for the Compustat matching process.

The Six Core BR Files are

1. Employer Identification Number (EIN): Unique identifier assigned by the IRS.
2. Parent-Entity EIN (EIN-851)[6]: Identifies the parent company using data from IRS Form 851 (applicable for specific years).
3. Standardized Firm Name: Constructed by combining variables "name1" and "name2" or using them individually.
4. Standardized Address: Includes Street address information derived from mailing or physical address variables.
5. City, State, and ZIP Code (CSZ): Extracted from mailing or physical address data.

[6]EIN-851 is a parent-entity EIN variable populated with values taken from IRS Form 851. For the EIN-851 file, firmid is obtained from the completed EIN file derived from CBPBR by matching on EIN and year.

6. Telephone Number: (when available).

Each of the six core BR files focuses on a specific "standardized matching variable" (like address or EIN) and stores unique records for every combination of that variable with the firm identifier ("firmid"), establishment identifier ("estabid"), and the active years for that specific combination within the BR. This "active" period is flagged using the CBPBR variable 'ACTV_STAT'.

For instance, the Standardized address file would encompass all unique combinations of the estabid, firmid, and standardized street address, along with the years present and years active for each combination.

By organizing the data in this manner, rather than maintaining separate annual BR files that combine all the aforementioned variables for each year, we can independently match different sets of Compustat information to BR information for all relevant years in a single join for each matching variable. Appendix A provides a detailed explanation of the procedure for creating these six independent BR files.

This method offers significant advantages over using separate annual BR files. First, it significantly reduces matching time. By organizing data by variable, we can independently match Compustat variables (name/address) to the BR for all relevant years in a single join for each matching variable instead of 45 different joins by year of data. This approach considerably reduces the time required for matching.[7] Additionally, this data structure minimizes storage space needed compared to maintaining separate annual files.[8] The files were generated from the CBPBR datasets (vintage v2020).[9]

2.3 Longitudinal Business Database Preparation

While the Business Register (BR) files utilized firmids generated from CBPBR data, the final GVKEY-firmid bridge relies on firmids from the updated LBD (LBDREV), vintage v2020.[10] Each BR record linked to a GVKEY-year Compustat record possesses an estabid.

[7]This procedure reduced the time required for matching by a factor of nearly 1,000. For earlier work with annual files, matching 16 years of Compustat data (2002-2017) required 248 hours (SAS "real time"), approximately 15.5 hours/year of data. Using the process described here, we matched 45 years of Compustat data (1976-2020) in 98 minutes, approximately 0.036 hours/year of data. Matching with annual files required approximately 600 SAS table creations per year, versus 2.6 with the more recent data design.

[8]For the earlier work, 14 years (2002-2015) of cleaned BR data with DQ Match codes required 122 Gb of disk storage, approximately 8.7 Gb/year of data. Our current independent BR files covering 45 years (1976-2020) require 75.3 Gb, or 1.67 Gb/year of data.

[9]The exceptions are EIN-851 and telephone number, which are not in the CBPBR files. EIN-851 is available in the BR "851" folders for 1992-2001 and BR einunits files for 2002-2019. Telephone number data is available in SSL MU and SU files for 1988-2001.

[10]Chow, Melissa, Fort, Teresa C., Goetz, Christopher, Goldschlag, Nathan, Lawrence, James, Perlman, Elisabeth, Stinson, Martha and White, T. Kirk, (2021), Redesigning the Longitudinal Business Database,

This estabid (along with the year) serves as a key to the corresponding LBD record for that establishment, containing the firmid assigned during LBD's longitudinal processing. Notably, LBD data on employment and NAICS codes also played a crucial role in both BR matching and subsequent de-duplication efforts.

LBD data played a crucial role in both refining the matching process and the final de-duplication stage. Specifically, employment and NAICS codes from the LBD were leveraged to enhance the accuracy of BR matching. We constructed a dataset from the LBDREV file containing unique combinations of year, firm identifier ("firmid"), and 4-digit NAICS code (spanning 1976-2020). This dataset focused on industry entries where the establishment's employment in a specific industry constituted at least 25% of its total employment for that year, and where payroll was positive. This enriched data then facilitated matching steps that combined the 4-digit NAICS codes with various combinations of name and address data (including street, city, state, and zip code). Furthermore, LBD data proved instrumental during the final de-duplication stage. This step aimed to identify the single, most accurate "firmid" among potentially multiple firmids linked to a specific combination of GVKEY and year.

3 Matching Business Register and Compustat

We matched companies between the Compustat and Business Register (BR) databases in multiple stages. Each stage focused on a specific set of variables (name, street address, city-state-zip, etc). Since the data for these variables resided in separate files within each database, the matching process was done sequentially for each variable set within a stage.

For example, to match on company name and address, we first match company names between Compustat and BR. Once we have that set of matches, we then match to addresses, ensuring the year in Compustat matches the timeframe of the name and address record in BR.

Since BR data are organized in independent files, stages requiring both name-address matching and other variables (like city, state, and zip code) can leverage previous results. The name-address matches from the initial stage can be reused for subsequent stages needing the same information. This approach, combining independent data structures and sequential matching, significantly sped up the process. This efficiency allowed us to use a wider range of matching criteria across multiple stages.

Below is a breakdown of these stages:

- Stage 1: Single pass. Matching solely on Employer Identification Number (EIN).
- Stage 2. Multiple passes. Matching on EIN and company name, street address, and various combinations of city/state/zip.
- Stage 3: Multiple passes. Matching NAICS code along with name, address, and city/state/zip combinations. Here, the thresholds for name and address similarity were less strict compared to stage 2.
- Stage 4: Single pass. Matching on phone number along with name and address, again with relaxed thresholds for name and address threshold compared to stage 2.

After the first 3 stages were completed, the matches found were then matched to the BR telephone data, and flagged as such (flag variable "mch_phone" = 1), and this flag was used as a match quality score during post-match de-duplication.

3.1 Linking Executive Information Data

This section details the process of creating a bridge between the Compustat ExecuComp database and the Longitudinal Employer-Household Dynamics (LEHD) database. By establishing this bridge, we can link ExecuComp data with firmid without using any company name or address information as aforementioned.

Compustat ExecuComp offers detailed compensation data for top executives at companies, sourced directly from their annual filings (SEC Form DEF 14A). This data, accessible by a company's unique identifier (GVKEY), includes information such as salary, age, and gender for each executive. Using PVS probabilistic name-matching[11] we assign a unique person identifier, or protected-identification-key (PIK), to roughly 70% of the managers listed in ExecuComp. This PIK acts as a common thread across Census datasets.

By matching ExecuComp data to the LEHD database using the PIK identifier, we were able to retrieve firmid (company identifier) and EIN (Employer Identification Number) from the LEHD for each matched employment record.[12] This established a critical link between ExecuComp's executive data, containing GVKEY for each executive, and firmid information in LEHD.

This newly created GVKEY-firmid bridge using the ExecuComp data serves dual roles. Firstly, it aids in de-duplication by enhancing the existing Compustat-BR matching process, by identifying and eliminating potential duplicate entries. Secondly, it expands the bridge by

[11] https://www.census.gov/about/adrm/linkage/projects/pvs.html

[12] The LEHD database tracks the place of work for each Person Identifier Key (PIK) on a quarterly basis. This place of work is identified by a combination of three identifiers: SEIN, EIN, and firmid.

incorporating previously unidentified GVKEY-firmid pairs from the Compustat-BR name and address matching phase.

The final matching rates achieved through these processes are summarized in Table 1. For a more granular view of matching success across different years, you can refer to Table 2. Additionally, Figure 2 provides a visual representation of these annual match rates.

Table 3 offers a tabulation of the matching status for the Stock Ownership Code variable across the 1976-2020 timeframe. The Stock Ownership Code variable in Compustat indicates a company's trading status. A code of "Stko=0" signifies a publicly traded company listed on major exchanges like NYSE, ASE, NASDAQ, or the Over-the-counter bulletin board. Conversely, "Stko=3" identifies companies that are publicly traded but not listed on a major exchange. These companies might be found on alternative exchanges or platforms like the Pink Sheets or other OTC markets. Finally, Table 4 details the total Compustat employment, revenue, and assets successfully matched to the LBD for the same period (1976-2020).

4 Cleaning and De-Duplicating Matches

Our goal here is to choose the "best" firmid(s) for each GVKEY-year pair. To choose the best match from a set of duplicates we looked at the characteristics that differentiate them from one another. To help ensure proper interpretation and use of these characteristics, we converted each to a "score" such that higher is better, all else being equal. We used two different categories of scores: match quality and match quantity.

Match quality scores encompassed various metrics, including string similarity (GEDN or Jaro-Winkler) for firm name and address, as well as a measure of 3-digit zip code agreement. Additional binary flags indicated whether matches included EIN or telephone number agreements for a given match pass, and another binary flag marked agreement on NAICS codes, with separate scores for 4-digit, 2-digit, and 1-digit levels. Furthermore, LBD-Compustat employment agreement for a matched firm ID was also considered. Match quantity scores were determined by the count of matched years for a given GVKEY-firm ID combination.

Our de-duplication process employs a multi-stage approach, leveraging the established match quality and quantity scores. At each stage, we progressively refine the selection criteria to identify the most suitable firmid for a given GVKEY-year pair.

Despite this staged approach, some GVKEY-year pairs may still have multiple qualifying firmids. This occurs when there isn't a single definitive "best" match based on the established scoring system. In these instances, we acknowledge the ambiguity and retain all qualifying firmids associated with the specific GVKEY-year pair for further analysis or manual

intervention.

4.1 Residual Duplication

It is important to acknowledge that researchers may encounter duplication (i.e., GVKEY-year attached to multiple firmids) in this bridge. About 25% of all GVKEYS in Compustat from 1976-2020 are matched to more than one FIRMID in a given year. This one-to-many can arise from the underlying data structure or de-duplication limitations. The former implies that in some instances, a single GVKEY-year pair within the Census data might legitimately correspond to multiple distinct firmids as in Figure 1. This reflects the inherent complexity of real-world business structures, where a single company in Compustat (representing a unique issue-currency-index) may operate under multiple Census's firm identifiers. The latter implies that our de-duplication process, while effective, may not definitively resolve all cases. This can occur when multiple firm IDs exhibit very similar characteristics, making it challenging to confidently select a single "best" match based on the established scoring criteria.

5 Conclusion

This paper details the creation of an updated bridge between Compustat and the Longitudinal Business Database (LBD) spanning 1976-2020. This comprehensive linkage facilitates enhanced analysis of firm behavior and performance by integrating detailed financial data across multiple decades. Our rigorous matching process, leveraging fuzzy name and address matching along with EIN, telephone number, and industry code, has resulted in a high match rate, ensuring the robustness and reliability of the dataset.

The incorporation of historical data from diverse sources, including CRSP, SEC filings, and ExecuComp, further enriches the dataset, providing a thorough representation of public companies over time. The creation of six distinct Business Register files focusing on specific matching variables significantly optimized the matching process, reducing both time and storage requirements. Additionally, the longitudinal aspect of the LBD data, with its detailed employment and NAICS codes, was instrumental in refining the matching and de-duplication stages.

Researchers now have access to a valuable resource that can facilitate a wide range of studies, from firm-level financial analysis to broader economic trends.

Figure 1: Relationship between GVKEYS and FIRMIDs

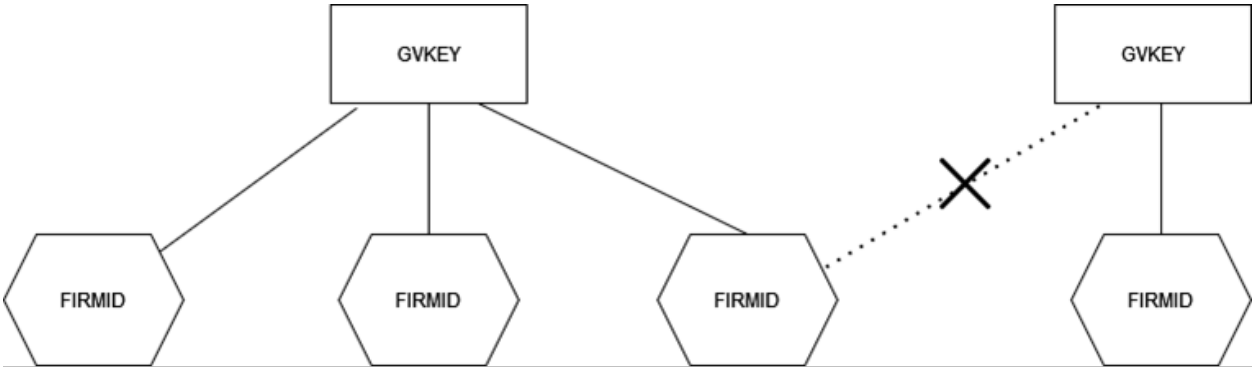


Table 1: Compustat Matching Rate, 1976–2020

	GVKEY Counts	**Percent**
Unmatched	48,000	15%
Matched	262,000	85%
Total	310,000	100%

Source: Compustat and Longitudinal Business Database. This table represents companies in Compustat (GVKEYs) that matched to LBD. GVKEYs in Compustat are restricted to fiscal years 1976–2020, U.S. location (loc=USA), excluding NAICS 525, and traded on major exchanges ($stko = 0$ or $stko = 3$).

Table 2: Compustat Matching Rate to LBD by Year, 1976-2020

Year	Unmatched GVKEYs	Matched GVKEYs	Total	Unmatched % Matched%/Total%
1976	2,200	3,500	5,800	38% / 60% / 100%
1977	2,100	3,700	5,800	36% / 64% / 100%
1978	2,000	3,800	5,800	34% / 66% / 100%
1979	2,000	3,900	5,900	34% / 66% / 100%
1980	1,800	4,100	5,900	31% / 69% / 100%
1981	1,600	4,500	6,000	27% / 75% / 100%
1982	1,400	5,000	6,400	22% / 78% / 100%
1983	1,100	5,500	6,600	17% / 83% / 100%
1984	950	5,800	6,800	14% / 85% / 100%
1985	1,000	6,100	7,000	14% / 87% / 100%
1986	1,000	6,300	7,300	14% / 86% / 100%
1987	1,000	6,400	7,400	14% / 86% / 100%
1988	1,000	6,400	7,400	14% / 86% / 100%
1989	950	6,500	7,400	13% / 88% / 100%
1990	900	6,600	7,600	12% / 87% / 100%
1991	950	6,900	7,900	12% / 87% / 100%
1992	900	7,400	8,400	11% / 88% / 100%
1993	950	7,900	8,900	11% / 89% / 100%
1994	900	8,200	9,100	10% / 90% / 100%
1995	1,000	8,500	9,500	11% / 89% / 100%
1996	950	8,700	9,600	10% / 91% / 100%
1997	850	8,500	9,400	9% / 90% / 100%
1998	900	8,400	9,300	10% / 90% / 100%
1999	950	8,200	9,100	10% / 90% / 100%
2000	850	7,800	8,600	10% / 91% / 100%
2001	850	7,200	8,100	10% / 89% / 100%
2002	850	6,900	7,700	11% / 90% / 100%
2003	850	6,600	7,400	11% / 89% / 100%
2004	800	6,300	7,100	11% / 89% / 100%
2005	800	6,100	6,900	12% / 88% / 100%

Continued on next page

Year	Unmatched GVKEYs	Matched GVKEYs	Total	Unmatched % Matched%/Total%
2006	800	5,800	6,600	12% / 88% / 100%
2007	800	5,500	6,300	13% / 87% / 100%
2008	800	5,200	6,000	13% / 87% / 100%
2009	800	5,000	5,900	14% / 85% / 100%
2010	900	4,900	5,800	16% / 84% / 100%
2011	1,000	4,900	5,900	17% / 83% / 100%
2012	1,100	4,900	6,000	18% / 82% / 100%
2013	1,100	4,800	5,900	19% / 81% / 100%
2014	1,000	4,700	5,700	18% / 82% / 100%
2015	950	4,600	5,500	17% / 84% / 100%
2016	950	4,400	5,300	18% / 83% / 100%
2017	900	4,200	5,100	18% / 82% / 100%
2018	950	4,000	5,000	19% / 80% / 100%
2019	900	3,800	4,700	19% / 81% / 100%
2020	800	3,400	4,200	19% / 81% / 100%
Total	48,000	262,000	310,000	15% / 85% / 100%

Source: Compustat and Longitudinal Business Database. This table represents companies in Compustat (GVKEYs) that matched to LBD. GVKEYs in Compustat are restricted to fiscal years 1976–2020, U.S. location (loc=USA), excluding NAICS 525, and traded on major exchanges ($stko = 0$ or $stko = 3$).

Source: Compustat and Longitudinal Business Database. This table represents companies in Compustat (GVKEYs) that matched to LBD. GVKEYs in Compustat are restricted to fiscal years 1976–2020, U.S. location (loc=USA), excluding NAICS 525, and traded on major exchanges ($stko = 0$ or $stko = 3$).

Figure 2: Matched vs Unmatched Compustat Firms (GVKEYs) 1976-2020

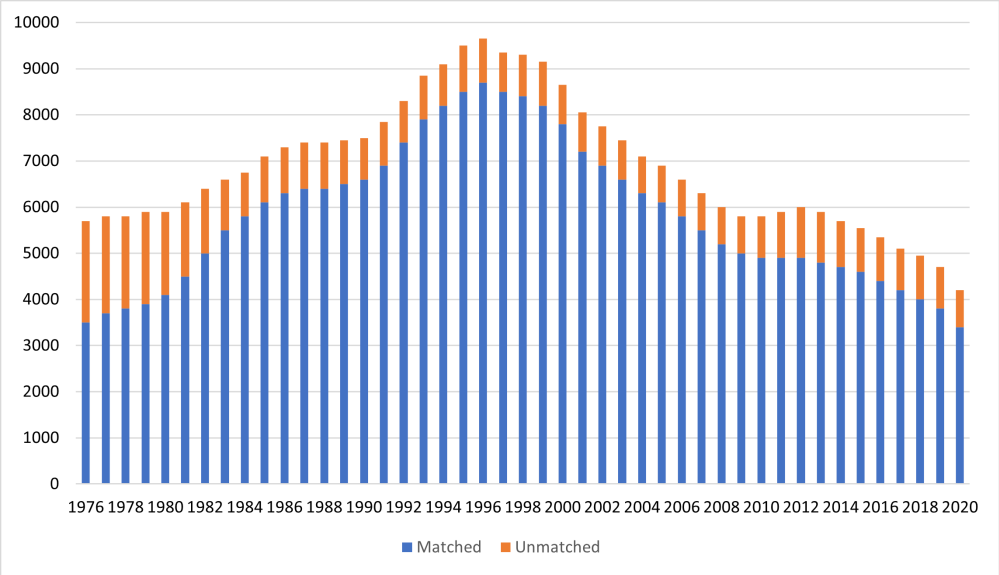


Table 3: Tabulation of Stock Ownership Code Match Status in Compustat, 1976–2020

Stock Ownership	**% Unmatched**	**% Matched**	**Total GVKEYs**
Publicly traded on major exchange ($stko = 0$)	13%	87%	221,000
Publicly traded on minor exchange ($stko = 3$)	21%	79%	88,500
Total			310,000

Source: Compustat and Longitudinal Business Database. Note: $stko = 0$ in Compustat denotes publicly traded company, includes NYSE, ASE, NASDAQ, and Over-the-counter bulletin board. $stko = 3$ in Compustat denotes Company that is publicly traded but not on a major exchange, includes Other, Pink Sheet, other OTC, etc.

Table 4: Percentage of Total Employment, Revenue, and Assets in Compustat Matched to LBD, 1976–2020

	% Unmatched	**% Matched**	**Total**
Employment	4.9%	95.1%	1,538,671
Revenue ($M)	5.0%	95.0%	379,868
Assets ($M)	4.5%	95.5%	1,019,034

Source: Compustat and Longitudinal Business Database. Note: This table shows the share of employment, revenue, and assets for GVKEYs in Compustat that were matched to the LBD. For the "Matched" column, the numerator in the first row represents the sum of employment across all matched GVKEYs in Compustat for the reference years. The denominator is the total employment in Compustat for the reference years. GVKEYs in Compustat are restricted to GVKEYs present in Compustat fiscal years 1976-2020. These GVKEYs have listed USA as location (loc=USA), are not NAICS 525 and are traded in major stock exchange, i.e., stko=0 or stko=3.

A Appendix A

To construct each of the six independent BR files, we performed the following steps:

- Create annual files from CBPBR, BR, or SSL files containing the match variable(s) and cleaned firmid and estabid (including both active and inactive firms).
- Combine all annual files by successively merging on the combination of the match variable (e.g., name or address), firmid and estabid, and then updating the set of BR annual flag variables indicating presence (active or inactive) or absence of each variable combination for that BR year.
- Standardize the match variable(s) (only for firm name, street address, city, and telephone number)
- Collapse on the standardized match variable, firmid, and estabid. (Some records with different non-standardized values have identical standardized values, and these need to be combined, with the annual flags updated to reflect all contributing data.)
- Create DQ Match codes using sensitivity 95 and appropriate data-type specification, for standardized firm name, street address, city. Reduce DQ Match code variable length to reduce file size.
- Determine first and last BR years present and first and last years active for each record.

B Appendix B

Below you will find instructions for the Company Organization Survey, taken from https://www2.census.gov/programs-surveys/cos/information-for-respondents/help/2016%20COS_NC-99001_Instructions_20170302_forHelpSite.pdf (Accessed September 19, 2025).

Definitions and Instructions for the Report of Organization, NC-99001

Estimates are acceptable if book figures are not readily available.

This survey contains questions about the organization, questions about each of the pre-identified locations of this organization (if any), and questions for any locations being added to this organization (if any).

Respondents will either have a short survey (S) or a long survey (L) to fill out. The (L) survey contains questions for the organization (survey NC-99001(L)), the pre-listed locations, and any additional locations to add. The (S) survey contains questions for the organization (survey NC-99001(S)), and any additional locations to add.

Respondents can fill out this survey via downloading a spreadsheet template or online.

- The spreadsheet template option requires downloading the spreadsheet from this instrument, filling out the spreadsheet, uploading the completed spreadsheet back to the instrument, reviewing and correcting the responses (as needed), and submitting the responses within this instrument.

- The online option is a question by question response driven approach. Respondents will navigate through this survey, based on responses to questions and clicking the 'Save and Continue' button, then review and correct the responses (as needed) and then submit these responses within this instrument.

Burden Statement

We estimate this survey will vary from a half-hour for the smallest companies to 25 hours for the largest companies to complete, with an average of 3.4 hours to complete, including the time for reviewing instructions, searching existing data sources, gathering and maintaining the data needed, and completing and reviewing the collection of information.

Send comments regarding this burden estimate or any other aspect of this collection of information, including suggestions for reducing this burden to: EWD Survey Comments 0607-0444, U.S. Census Bureau, 4600 Silver Hill Road, Room EWD-8K154, Washington, DC 20233. You may e-mail comments to EWD.surveys@census.gov. Be sure to use "EWD Survey Comments 0607-0444" as the subject.

Response to this collection of information is not required unless it displays a valid approval number from the Office of Management and Budget (OMB). The OMB eight-digit number appears in the upper right corner of the instrument.

Discontinued Operations - If your organization discontinued operations before 2016, complete the Operational Status section of the questionnaire for each location (e.g., Closed or Sold) along with the effective date for of the status change. If your organization was in operation for any part of 2016, a completed questionnaire is required for the portion of the year 2016 in which your organization was in operation.

Census File Number (CFN) - A CFN has been assigned to your company. Always refer to the complete number in any correspondence with the U.S. Census Bureau.

Employer Identification Number (EIN) - The EIN is the nine-digit taxpayer identification number assigned by the Internal Revenue Service (IRS) and used by all business firms and other legal entities to file federal employment tax returns such as Forms 941 and 943; and federal income tax returns such as Forms 1065, 1120, and 990 series.

Company - An economic unit comprising one or more establishments under common ownership or control.

Subsidiary - A company that is owned or controlled by another firm or company. Subsidiaries include firms in which your company owns more than 50 percent of the outstanding voting stock, as well as firms in which your company has the power to direct or cause the direction of the management and policies.

Establishment - An establishment is a single physical location where business is conducted or where services or industrial operations are performed. (For example: a factory, mill, store, hotel, movie theater, mine, farm, and administrative office.) If distinctly separate activities are performed at the same location, and if there is significant employment in each activity, then each activity should be treated as a separate establishment, except agricultural production as explained below. (For example: construction activities operated out of the same physical location as a lumber yard.)

For firms engaged in construction, real estate management or operation, transportation, communication, electric, gas, and sanitary services, and similar types of physically dispersed activities, it is not necessary to list separately each individual "site," "project," "field," "network," "line," or "system." It is only necessary to report for main or branch offices, terminals, stations, etc., which are either (a) directly responsible for supervising such activities, or (b) the base from which personnel operate to carry out these activities.

Foreign Locations - Report a foreign location only if any of the employees at the location are reported on an IRS Form 941 for an EIN that is also used to report employees at domestic locations. You only need to report one foreign location for each such EIN. If more than one such establishment exists for an EIN, consolidate the employment and payroll data at the headquarters location, using "foreign employees" as the secondary name.

Mailing Address

This information is what the U.S. Census Bureau has on file for this establishment and lists the location this survey was mailed to. Respondents may update ATTN, Name 1, Name 2, Street, City, State, and ZIP Code fields if the information displayed is incorrect.

Ownership or Control: Voting stock validation

Select the suitable answer to indicate if another domestic company owns more than 50 percent of the voting stock of your company.

Ownership or Control: Management and policy

Select the suitable answer to indicate if another domestic company has the power to control the management and policies of your company.

Ownership or Control: Percent of Voting Stock Held

Select the suitable answer to indicate the percentage of voting stock held by the owning or controlling company. Franchisees should not list their franchisor as a controlling company unless they own more than 50 percent of the voting stock.

Ownership or Control: Company Information

Enter the name, home office address, and Employer Identification Number (EIN) of the owning or controlling company. Franchisees should not list their franchisor as a controlling company unless they own more than 50 percent of the voting stock.

Foreign Ownership or Control

Select the suitable answer to indicate if a foreign entity (company, individual, government, etc.) owns directly or indirectly 10 percent or more of the voting stock or an equivalent interest in your company.

Foreign Ownership or Control: Company Information

Enter the name, address, and country of the foreign entity (company, individual government).

Foreign Ownership or Control: Percent of Voting Stock Owned

Select the suitable answer to indicate the percentage of voting stock owned directly or indirectly by a foreign entity (company, individual, government).

All direct and indirect ownership interests held by the foreign entity should be summed to determine the foreign entity's percentage of ownership.

Foreign Affiliates

Select the suitable answer to indicate if your company owns 10 percent or more of the voting stock or other equity rights of a foreign business enterprise (including an unincorporated branch, partnership, or ownership of real estate).

Research and Development

Select the suitable answer to indicate if your company performed or funded research and development for the year.

Research and Development (R&D) is planned, creative work aimed at discovering new knowledge or developing new or significantly improved goods and services.

This **includes**:

- activities aimed at acquiring new knowledge or understanding without specific immediate commercial applications or uses (basic research)
- activities aimed at solving a specific problem or meeting a specific commercial objective (applied research)
- systematic use of research and practical experience to produce new or significantly improved goods, services, or processes (development)

The term research and development **excludes**:

- routine product testing, quality control, and technical services unless they are an integral part of an R&D project
- market research
- efficiency surveys or management studies
- literary, artistic, or historical projects
- prospecting or exploration for natural resources

Research and Development Expenses

Select the suitable answer to indicate your company's worldwide expenses for research and development for the year.

Professional Employer Organization

Select the suitable answer to indicate if your company leases fifty percent or more of its permanent full- and part-time workforce through a contractual agreement with a Professional Employer Organization,

Permanent workforce excludes contractors and temporary staffing from a staffing service.

Pre-listed Locations

This section of the survey contains questions for all the pre-listed locations of the organization.

A. Location Information

For each location, please update the following information as needed: major activity, Name, Secondary Name, Store or plant No. Physical Location (number and Street), City, State, and ZIP

Discontinued Operations

If your organization discontinued operations before the survey year, please select "Ceased Operation" and select a date for "CEASED OPERATION OR SOLD OR LEASED INFORMATION" for each Prelisted location. Return to the Remarks section with an annotation that your organization did not operate in any part of the survey year. If your organization did operate in any part of the survey year, we require a report for the portion of the survey year in which your organization was in operation.

Special Instructions for Selected Industries

Agricultural production

Establishments having any agricultural production regardless of whether it is the major activity should be listed separately and the employment and payroll data allocated if appropriate. Agricultural production includes the raising of/or production of crops (including horticultural products), poultry, or livestock for sale.

Construction establishment

A construction establishment is a relatively permanent office, or other place of business, where the usual business activities related to construction are conducted. With some exceptions, a relatively permanent office is one which has been established for the management of more than one project or job and which is expected to be maintained on a continuing basis.

Banks

When checking the pre-listing of your establishments, list only the EINs used to report either Federal employment taxes (IRS Form 941) or income (IRS Forms 1065, 1120, and 990). DO NOT LIST EINs assigned to trust funds or estates managed by the bank and reported on IRS Form 1041, U.S. Fiduciary Income Tax Return. Report main and branch locations as separate establishments. Individual ATM outlets should not be listed as separate establishments.

Retail departments or concessions

List as a separate establishment (including specific street address) those retail or service outlets operated by your firm as a department of a retail or service establishment, such as shoe departments or beauty salons in department stores, meat or bakery departments in grocery stores, or millinery departments in clothing stores.

Medical

Part-time offices should not be considered as separate establishments. List only the offices that are staffed on a full-time basis and include the employment and payroll for the part-time offices. Home health care services should not consider the customers' locations as separate establishments. Employment and payroll for these locations should be included with the main or branch office from which the work is supervised.

Legal

Part-time, temporary, or special offices should not be considered separate establishments for purpose of this report. Data for such locations should be included with the controlling main or branch location where the service is primarily performed.

B. Employment and Payroll

Enter the number of employees for the pay period including March 12th, the first quarter payroll, and the annual payroll for the location for the survey year.

Employment:

Please use the number of employees, both full- and part-time whose payroll was reported on your organization's Internal Revenue Service Form(s) 941, Employer's Quarterly Federal Tax Return. Include as employees all persons on paid sick leave, paid holidays, and paid vacations; include salaried officers and executives of incorporated firms.

Exclude full- or part-time leased employees whose payroll was filed under an employee leasing company's EIN and temporary staffing obtained from a staffing service.

Payroll

Include the total of wages paid, tips reported, and other compensation paid to your employees in the calendar year, whether or not subject to income or social security taxes. Include all wages, salaries, commissions, fees, bonuses, vacation allowances, sick leave pay, severance pay, the amount of reported tips, and the value of taxable fringe benefits. Also, include any employee contributions to a qualified pension plan, such as the 401(k) plan. Also, include the spread on stock options that are taxable to employees as wages.

For incorporated businesses, include the salary of all officers and executives.

Exclude pensions, annuities, supplemental unemployment compensation benefits, nontaxable fringe benefits, and commissions and fees paid to independent contractors. Also, exclude employer contributions for pension plans, health insurance, and qualified benefits under a cafeteria plan. For unincorporated businesses, exclude the profit or compensation paid to proprietors or partners.

The sum of payroll reported for individual establishments for an EIN should equal the taxable Medicare wages and tips reported on Item 5c of IRS form 941, Employer's Quarterly Federal Tax Return plus the spread on stock options that are taxable to employees.

C. Operational Status

Select the box that is applicable to the operation of your location during the survey year.

- If the establishment ceased operation,
  - Enter the ceased operation date in the **CEASED OPERATION OR SOLD OR LEASED INFORMATION** response section.

- If the establishment sold or leased to another operator,
  - Enter the sold operation date in the **CEASED OPERATION OR SOLD OR LEASED INFORMATION** response section
  - Also, enter the name and address of this establishment's new owner or operator.

Added Locations

This survey contains questions for locations being added to the organization.

Do not add locations from the pre-listed locations section, or plants for which you received an MA-10000 survey.

A. Location Information

For each new establishment being added, enter the EIN, name, secondary name, Store/plant No., Physical location, City, State, ZIP, and date establishment opened or is expected to open.

Special Instructions for Selected Industries

Agricultural production

Establishments having any agricultural production regardless of whether it is the major activity should be listed separately and the employment and payroll data allocated if appropriate. Agricultural production includes the raising of/or production of crops (including horticultural products), poultry, or livestock for sale.

When adding establishments engaged in agricultural production, be sure to give the type of crops, livestock, etc.

Construction establishment

A construction establishment is a relatively permanent office, or other place of business, where the usual business activities related to construction are conducted. With some exceptions, a relatively permanent office is one which has been established for the management of more than one project or job and which is expected to be maintained on a continuing basis.

Banks

When adding establishments, list only the EINs used to report either Federal employment taxes (IRS Form 941) or income (IRS Forms 1065, 1120, and 990). DO NOT LIST EINs assigned to trust funds or estates managed by the bank and reported on IRS Form 1041, U.S. Fiduciary Income Tax Return. Report main and branch locations as separate establishments. Individual ATM outlets should not be listed as separate establishments.

Retail departments or concessions

List as a separate establishment (including specific street address) those retail or service outlets operated by your firm as a department of a retail or service establishment, such as shoe departments or beauty salons in department stores, meat or bakery departments in grocery stores, or millinery departments in clothing stores.

Medical

Part-time offices should not be considered as separate establishments. List only the offices that are staffed on a full-time basis and include the employment and payroll for the part-time offices. Home health care services should not consider the customers' locations as separate establishments. Employment and payroll for these locations should be included with the main or branch office from which the work is supervised.

Legal

Part-time, temporary, or special offices should not be considered separate establishments for purpose of this report. Data for such locations should be included with the controlling main or branch location where the service is primarily performed.

B. Employment and Payroll

Enter the number of employees for the pay period including March 12th, the first quarter payroll, and the annual payroll for the location for the survey year.

Employment:

Please use the number of employees, both full- and part-time whose payroll was reported on your organization's Internal Revenue Service Form(s) 941, Employer's Quarterly Federal Tax Return. Include as employees all persons on paid sick leave, paid holidays, and paid vacations; include salaried officers and executives of incorporated firms.

Exclude full- or part-time leased employees whose payroll was filed under an employee leasing company's EIN and temporary staffing obtained from a staffing service.

Payroll

Include the total of wages paid, tips reported, and other compensation paid to your employees in the calendar year, whether or not subject to income or social security taxes. Include all wages, salaries, commissions, fees, bonuses, vacation allowances, sick leave pay, severance pay, the amount of reported tips, and the value of taxable fringe benefits. Also, include any employee contributions to a qualified pension plan, such as the 401(k) plan. Also, include the spread on stock options that are taxable to employees as wages.

For incorporated businesses, include the salary of all officers and executives.

Exclude pensions, annuities, supplemental unemployment compensation benefits, nontaxable fringe benefits, and commissions and fees paid to independent contractors. Also, exclude employer contributions for pension plans, health insurance, and qualified benefits under a cafeteria plan. For unincorporated businesses, exclude the profit or compensation paid to proprietors or partners.

The sum of payroll reported for individual establishments for an EIN should equal the taxable Medicare wages and tips reported on Item 5c of IRS form 941, Employer's Quarterly Federal Tax Return plus the spread on stock options that are taxable to employees.

C. Major Activity Code

Select the Activity Code that best describes the activity of this establishment and specify the principal products or services.

D. Former Owner or Operator Information

If this added location was acquired from a former owner, enter the name of the former owner, the mailing address (Number and Street), City, State, ZIP, and the date this location was acquired.

References

Chow, Melissa, Fort, Teresa C., Goetz, Christopher, Goldschlag, Nathan, Lawrence, James, Perlman, Elisabeth, Stinson, Martha and White, T. Kirk, (2021), Redesigning the Longitudinal Business Database, Working Paper Series 21-08, U.S. Census Bureau, Center for Economic Studies.